ADEMP-PreReg
Template for Simulation Studies

October 30, 2026

Version: 2.0
Last updated: 2026-10-30
Template based on the papers:
Angelike, T., & Heck, D. W. (2026). Evaluating Ratings from Large Language Models: A Simulation-Based Workflow. https://osf.io/preprints/psyarxiv/xphm9
and
Siepe, B. S., Bartoš, F., Morris, T. P., Boulesteix, A.-L., Heck, D. W., & Pawel, S. (2024). Simulation Studies for Methodological Research in Psychology: A Standardized Template for Planning, Preregistration, and Reporting. *Psychological Methods*. https://doi.org/10.1037/met0000695, https://doi.org/10.31234/osf.io/ufgy6

Instructions

General Information

This template treats the repeated generation of ratings from one or more large language models (LLMs) as a simulation study (for details, see Angelike & Heck, 2026). It adapts the ADEMP-PreReg template for Monte Carlo simulation studies (Siepe et al., 2024). A Word version is available at https://github.com/TImA97/ADEMP-PreReg-for-LLMs. The LaTeX version provides simple commands to exclude the instructions, explanations, and example boxes from the compiled document. To preregister and time-stamp your protocol, we recommend uploading it to the Open Science Framework (https://osf.io/) or Zenodo (https://zenodo.org/). When using this template, please cite the associated articles (Angelike & Heck, 2026; Siepe et al., 2024). If you have any questions or suggestions for improving the template, please contact us via the ways described at https://github.com/TImA97/ADEMP-PreReg-for-LLMs.

Using this template

This template can be used for any study that uses LLM prompting to annotate, classify, or rate text materials (e.g., regarding psychological, semantic, or syntactic properties). Such applications involve many degrees of freedom: different task instructions, prompt details, and model versions may lead to different LLM output, and even repeated runs of an identical prompt may yield different results. The template therefore evaluates the quality of LLM ratings as a function of these (often arbitrary) decisions concerning prompt design, model choice, and implementation (e.g., whether ratings are taken from the generated text or derived from the probability distribution over response-option tokens). It can be adapted if ratings are obtained using other natural language processing methods (e.g., semantic vector embeddings, fine-tuned encoder classifiers).
Please provide detailed answers to each of the questions. If you plan to perform multiple LLM simulation studies within the same project, you can either register them separately or number your answers to each question with an indicator for each study. As the planning and execution of LLM simulation studies often involves considerable complexity and unknowns, it may be difficult to answer all the questions in this template or some changes may be made along the analysis pathway. This is to be expected and should not deter researchers from planning or preregistering an LLM simulation study. If you preregistered your LLM simulation, any modifications to the protocol should simply be reported transparently along with justification, which will ultimately add credibility to your research.

1 General Information

1.1 What is the title of the project?

Example

Evaluating the quality of LLM ratings for the valence and arousal of multi-word expressions

Answer:

1.2 Who are the current and future project contributors?

Example

Tim Angelike and Daniel Heck

Answer:

1.3 Provide a description of the project.

Explanation: This can include the overarching research questions and goals as well as empirical data that will be analyzed within the same project. If you are planning to conduct not only a simulation study but also an empirical study with human raters to validate the LLM output, we recommend preregistering the empirical study separately (e.g. regarding the study design and how to deal with outliers or missing data).

Example

We will investigate the quality of LLM-generated ratings of valence and arousal regarding multi-word expressions curated by Martínez et al. (2024). In two separate simulation studies, we will vary the specific LLM and the prompt structure composed of the required response format and the examples provided to the models as context. Our analysis will focus on (1) a variance decomposition of LLM-generated ratings and (2) the correlation of LLM and human ratings of valence and arousal. Human-generated ratings of valence and arousal were also provided by Martínez et al. (2024). We plan to report only the results for valence in the main body of the text, whereas the results for arousal will be reported in the supplementary material.

Answer:

1.4 Did any of the contributors already conduct related simulation studies on this specific question?

Explanation: This includes preliminary LLM prompting in the context of the current project.

Example

Before conducting the simulation, preliminary tests were run on ratings of syntactic complexity curated by Schmidt and Heck (2024) to ascertain the functionality of the analysis pipeline.

Answer:

2 Aims

2.1 What is the aim of the LLM simulation study?

Explanation: The aim of an LLM simulation study refers to the goal of the research and shapes subsequent choices. Aims are typically related to evaluating the capability of different LLMs to fulfill a particular task like generating ratings for word lists. Depending on whether an external validation criterion is available to evaluate the LLM output, one can pursue one or both of two types of aims:

1. Investigating variance components in LLM-generated ratings to distinguish: (1) *construct variance* reflecting stimulus differences, (2) *method variance* introduced by factors such as model choice or prompt design, and (3) *residual variance* due to repetitions of LLM calls (i.e., stochastic error). Potential aims include a variance decomposition of LLM ratings regarding different models and prompting approaches, or the assessment of inter-rater reliability between different LLMs or between the stochastic, repeated output of the same LLM. Note that high consistency does not imply correct or valid ratings; LLMs may agree while being systematically biased.

2. Investigating the correspondence of ratings obtained from one or more LLMs with an external gold standard, e.g., human ratings for the same task. This aim is most closely related to the traditional idea of simulation studies where a statistical method is evaluated with respect to its performance in recovering known, true parameters. Depending on whether the target is subjective (e.g., human norms) or objective (e.g., factual truth), relative or absolute agreement may be of interest.

Example

The aims of the present study are (1) to conduct a variance decomposition of LLM ratings of valence and arousal regarding multi-word expressions into construct variance (due to differences in the valence and arousal of the included expressions), method variance (due to specific LLM and prompt variations), as well as residual variance (due to repeated LLM calls), and (2) to assess the extent to which LLM ratings correspond to the perceived valence and arousal of multi-word expressions as rated by humans.

Answer:

3 Data-Generating Mechanism

3.1 Which materials will be used as input for LLM prompting?

Explanation: The stimulus materials used for LLM prompting (i.e., the corpus) can include statements, questions, vignettes, or items from a personality inventory. The specific set of input materials determines the scope and the generalizability of the observed findings regarding the quality of LLM-generated ratings. If the corpus consists only of statements assessing a particular psychological construct, the results of the LLM performance evaluation are informative only for comparable materials and constructs. Hence, it should be ensured that the verbal materials used as input are sufficiently broad and heterogeneous to increase the generalizability of conclusions. However, what constitutes sufficiently diverse data as input depends on the research question. Describe how items were sampled from the target domain. When the goal is to rate a large corpus of stimuli using LLMs, a random subset can be used for the validation study.

Example

Our study uses two different sets of $n_{\text{items}} = 96$ English multi-word expressions developed by Martínez et al. (2024). The first set of multi-word expressions was rated regarding their valence by human raters in Experiment 4 of Martínez et al. (2024). The second set of multi-word expressions was rated regarding their arousal by human raters in Experiment 5 of Martínez et al. (2024). They selected the stimuli based on their own GPT-4o ratings to cover the entire range of the respective construct. All analyses will be conducted separately for the multi-word expressions of valence and arousal but with identical methodology reported in the following.

Answer:

3.2 Which procedures will be used to obtain LLM output?

Explanation: LLM ratings should be generated in an automated and standardized manner. Describe all procedural details that are identical across all simulation conditions. Specify only the general structure of the LLM prompt here but report specific details of the prompt design that vary across conditions in Section 5 (Methods). Clearly state what type of software is used to obtain LLM ratings (e.g., a closed API, an open-weights model run locally, or access through a chat interface). Furthermore, state whether and how a specific response format was ensured, for instance, by including specific instructions in the prompt or requiring LLM output to have a specific format such as JSON. Note that whether open or structured output is used itself constitutes a researcher degree of freedom. We generally recommend requesting structured output to increase standardization, and using tools designed to process it (e.g., the `type_object()` function family in the `ellmer` package in R; Wickham et al., 2025). To ensure independence between ratings, it is important to use a fresh context when eliciting LLM responses (Conde et al., 2026).

Example

We will use open-weights LLMs which are run locally through the software `Ollama` and are accessed via the R package `ellmer`. To ensure that LLM ratings of valence and arousal only consist of numerical values, we use the `type_object()` and `type_integer()` functions to request output in JSON format with a predetermined response format of integers. Each prompt is composed according to the structure used by Martínez et al. (2024) in prompt 2 (valence) and prompt 3 (arousal) of their paper: (1) An explanation of the task, (2) the range of responses allowed, (3) examples for multi-word expressions, (4) the target multi-word expression to be rated, and (5) a reminder for the range of responses allowed. Each LLM call will start with a fresh context (no conversation history). A pre-generated random seed will be passed with each request.

Answer:

4 Estimands and Targets

4.1 What will be the estimands and/or targets of the LLM simulation study?

Explanation: Estimands and other targets refer to the practical aims of the compared methods. In contrast to traditional simulation studies (see Siepe et al., 2024), the estimands in an LLM simulation study are not certain parameters of a statistical method, but rather the (unknown) values of the stimuli on the variables to be extracted from text data. Please also specify if some targets are considered more important than others, i.e., if the simulation study will have primary and secondary outcomes.

Example

Our target estimands are the valence and arousal of multi-word expressions (where the scale varies across conditions; see Section 5.2). Valence refers to the pleasantness of a stimulus (very negative to very positive), arousal to the intensity of emotional activation it elicits (calm to excited).

Answer:

5 Methods

5.1 Which LLMs will be used to generate ratings?

Explanation: LLMs differ in numerous aspects, e.g., the architecture, the number of parameters, the company or institution developing a model, the language in which they were trained, and whether model weights and/or information about the training procedure are publicly available. State which LLMs will be used. Report the values of all hyperparameters that may affect the output, including those left at their defaults, and verify the settings actually in effect. Variability across iterations (residual variance) is

only interpretable if temperature and other sampling parameters are fixed, reported, and ideally equated across models. If hyperparameters are of interest, they can be varied as separate method factors, such as the temperature, top-p, top-k, the context length, and the batch size. If possible, include information on why you selected the models, which can concern aspects such as availability, expected costs, openness, and/or computational feasibility. A fully factorial design including all combinations of LLM model families and different model sizes may often be infeasible, and thus we recommend including all investigated LLM model versions as a single factor in the analysis (as opposed to handling model family and model size as separate factors).

Example

We include eight open-weights models by four companies that can efficiently be run locally using Ollama on a GPU with 32GB VRAM without requiring access to an online service. The eight models are gemma3 (with 4b, 12b, and 27b parameters; Google), llama3.1 (8b parameters; Meta), phi3 (3.8b, 14b parameters; Microsoft), phi4 (14b parameters; Microsoft), and gpt-oss (20b parameters; OpenAI). All models were updated to their most recent version as of 28 January 2026. All parameters will be kept at the values that apply by default in our software setup (Ollama 0.12.9 accessed via ellmer 0.4.0), which results in temperature = 1 and top-p = 1 for all models and top-k = 64 (gemma3) or 40 (other models). All other parameters remain at the Ollama or model defaults, including a context length of 4,096 tokens (num_ctx), which is far longer than our short prompts (8,192 for gpt-oss due to Ollama's settings). Requests will be sent one at a time (no parallel requests); Ollama's num_batch (prompt tokens per forward pass) remains at its default of 512.

Answer:

5.2 How will LLM prompts be structured and formulated?

Explanation: Be as specific as possible regarding the procedure that will be used to construct LLM prompts. Include information about specific parts of the prompts, such as system prompts that may contain context about the purpose of the task, instructions for the LLM on which scale to return ratings, and whether examples are provided or not (zero-shot vs. few-shot prompting). Depending on the research question, a fully factorial design or a single factor containing all distinct prompt versions may be appropriate. To ensure that the comparison is meaningful, the same prompt versions should be used for all LLM versions. Prompts should be preregistered and reported verbatim rather than paraphrased. This includes (a) the exact prompt strings for all conditions, including whitespace, line breaks, and special characters, (b) the system prompt or, if none was used, a statement that it was left empty, (c) the output schema or format instructions passed to the model (e.g., a JSON schema), and (d) how prompts were constructed, with a placeholder marking where the stimulus is inserted. The most reliable documentation is to share the prompts as plain-text files together with the code that assembles them.

Example

We adopt the prompt design by Martínez et al. (2024) to obtain valence and arousal ratings separately while manipulating two key components in a fully factorial setting:

1. The format of the rating task (a scale from 1 to 9 vs. a scale from 0 to 100) and
2. The presence and type of example statements provided: no example expressions; three examples for low valence or arousal (lower anchor); three examples for high valence or arousal (upper anchor); or three examples for both low and high valence or arousal (both lower and upper anchor). The last condition corresponds to the original prompt design used in Martínez et al. (2024).

Ratings of valence and arousal will be obtained in two separate and independent steps of the LLM simulation. An empty system prompt will be passed; no custom system instructions will be provided. Example prompts can be found at the end of this document in the Appendix.

Answer:

6 Performance Measures

6.1 Which performance measures will be used?

Explanation: Please state why specific performance measures are used and how they will be calculated. You may provide formulas to avoid ambiguity. The choice for specific performance measures strongly depends on the aim of the investigation. If the aim is to analyze the proportion of variance in LLM ratings due to different LLM model versions or prompt instructions, it may be necessary to specify a suitable mixed-effects model for the variance decomposition (e.g., using generalizability theory; Brennan, 2001). If an external criterion is available against which LLM output is compared (e.g., human ratings), different measures of inter-rater reliability or agreement between human and LLM ratings can be computed (e.g., the correlation of ratings across all stimulus materials). In some scenarios, it may also be of interest to compare LLM output to an established gold standard using the bias or RMSE (e.g., when assessing whether LLMs can accurately predict a certain outcome of interest).

Example

The present investigation has two aims:

1. Estimating variance components in LLM valence and arousal explained by differences in the stimulus materials and differences in methods (secondary performance measure) using generalizability theory (Brennan, 2001).
2. Assessing the correlation between human and LLM ratings of valence and arousal (primary performance measure).

All analyses will be conducted separately for expressions rated regarding their valence and the expressions rated regarding their arousal.

Regarding aim (1), we will estimate the variance components of the following linear mixed model using the `gtheory` package (Moore, 2016) which uses the `lmer()` function in R,

- `lmer(llm_rating ~(1 | stimulus * model * format * example), ...)`

The "stimulus" term accounts for differences in the target construct between the expressions (i.e., actual differences in valence or arousal), whereas the remaining terms ("model", "format", "example") account for methods effects (the facets in generalizability theory). If the mixed model cannot be fitted due to convergence issues, we will simplify the model structure by iteratively dropping higher-order interactions (first, the four-way interaction, then the three-way interactions not including the "stimulus" term and so on).
In line with generalizability theory, we will first perform a G study. We will fit the mixed-effects model above to estimate the variance components associated with each random intercept (main effects and interactions; Brennan, 2001).
In the D study, the universe score variance is defined as $\sigma^2(\tau) = \sigma^2(\text{stimulus})$ (i.e., as the random-intercept variance of items), because we aim to evaluate how well the obtained stimulus LLM ratings generalize across the investigated facets of the design (8 models, 4 example conditions, 2 response formats). The D study yields two types of coefficients. Our analysis will focus on the generalizability coefficient $E\rho^2$, which compares the universe score variance to the relative error variance $\sigma^2(\delta)$. The relative error consists of all components that include an interaction with the stimulus term, and therefore affect the ordering of stimulus ratings:

$$E\rho^2 = \frac{\sigma^2(\tau)}{\sigma^2(\tau) + \sigma^2(\delta)} \tag{1}$$

Similar to consistency measures such as Cronbach's α, $E\rho^2$ ranges from 0 to 1, with higher values indicating stronger agreement in the ordering of stimuli across facets. For completeness, we also report the dependability coefficient Φ, which compares the universe score variance variance against the absolute error variance $\sigma^2(\Delta)$ and is defined as

$$\Phi = \frac{\sigma^2(\tau)}{\sigma^2(\tau) + \sigma^2(\Delta)} \tag{2}$$

However, we do not consider the dependability coefficient to be relevant for the present investigation, as our study focuses on whether LLMs can produce stimulus ratings that show the same relative ordering (but not necessarily the exact same values) as those provided by humans.

Example

Regarding aim (2) we will compute the correlation between human and LLM ratings of valence and arousal using the Pearson correlation coefficient,

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x}) \cdot (y_i - \bar{y})}{(n-1) \cdot \mathrm{SD}(x) \cdot \mathrm{SD}(y)}. \tag{3}$$

We will compute the correlation at different levels of the analysis:

- (a) On the level of individual iterations in each simulation condition, we will compute the correlation between human ratings and LLM ratings across expressions. The inter-rater reliability (IRR) will be estimated by the average correlation:

$$\widehat{IRR} = \frac{\sum_{i=1}^{n_{\text{sim}}} \hat{r}_i}{n_{\text{sim}}}. \tag{4}$$

  To assess the estimation error due to the finite number of iterations, we will compute the Monte Carlo standard error:

$$\mathrm{MCSE}_{\widehat{IRR}} = \frac{\mathrm{S}_{\widehat{IRR}}}{\sqrt{n_{\text{sim}}}} \tag{5}$$

  where $\mathrm{S}_{\widehat{IRR}} = \sqrt{\sum_{i=1}^{n_{\text{sim}}}\{\hat{r}_i - (\sum_{i=1}^{n_{\text{sim}}} \hat{r}_i / n_{\text{sim}})\}^2/(n_{\text{sim}}-1)}$ is the sample standard deviation of the estimate and n_{sim} is the number of simulation iterations (see Siepe et al., 2024).

- (b) On the level of aggregated LLM ratings (averaged across all simulation iterations in each simulation condition), we will compute the correlation between human and LLM ratings. Using a bootstrap procedure, n_{est} LLM ratings (i.e., the number of ratings averaged per stimulus, e.g., 1–100) are first randomly drawn with replacement from the n_{sim} iterations of a simulation condition for each stimulus and then aggregated into an average rating. These average ratings per stimulus will then be used to compute the inter-rater reliability between human-generated and LLM-generated ratings. This procedure is repeated n_{sim} times to ensure that the same number of estimates is obtained as when using raw LLM ratings.

Answer:

6.2 How will the uncertainty in the estimated performance measures be calculated and reported?

Explanation: The estimation uncertainty of some performance measures (such as the correlation of LLM and human ratings) is bounded by the number of stimulus items n_{items}

and can be assessed by averaging the corresponding standard errors across simulation iterations. A different source of uncertainty is due to the stochasticity of repeated LLM ratings across iterations, which can be reported in the form of the Monte Carlo standard error (MCSE). Please see Siepe et al. (2024) and Morris, White, and Crowther (2019) for a list of formulae to calculate the MCSE related to common performance measures.

Example

We will average the standard error for the estimated correlation of LLM and human ratings for the $n_{\text{items}} = 96$ expressions across iterations. This accounts for uncertainty due to the limited number of expressions in the study. We will also compute Monte Carlo standard errors (MCSEs) using the formula by Siepe et al. (2024). This accounts for uncertainty regarding the limited number of simulation iterations in each condition. We will report the different types of standard errors and corresponding 95% confidence intervals in tables and in plots.

Answer:

6.3 How many simulation repetitions will be used for each condition?

Explanation: Please also indicate whether the chosen number of simulation repetitions is based on sample size calculations, on computational constraints, rules of thumb, or any other heuristic or combination of these strategies. Formulas for sample size planning in simulation studies are provided in Siepe et al. (2024). If there is a lack of knowledge on a quantity for computing the Monte Carlo standard error (MCSE) of an estimated performance measure (e.g., the variance of the estimator is needed to compute the MCSE for the bias), pilot simulations may be needed to obtain a guess for realistic/worst-case values.

Example

We will perform 100 simulation iterations per condition. The number of iterations was determined based on preliminary testing with ratings of syntactic complexity (Schmidt & Heck, 2024), which showed that this number of iterations resulted in narrow MCSE 95% confidence intervals while ensuring computational feasibility.

Answer:

6.4 How will invalid and missing LLM output be handled?

Explanation: LLM output can deviate from the allowed range of responses, resulting in invalid or missing ratings. If only numerical values in a predefined range are admissible, provide information on how other output (e.g., non-numeric text) will be treated. One possibility is to handle these values as missing data and report their relative occurrence rate. If a specific format has not been enforced by JSON format or other means, it is possible that some of the output may be in an open text format from which the information of interest can potentially be retrieved by further text processing. In this example case,

describe the steps taken to extract relevant information from the open text format. Alternatively, it is possible to impute or exclude invalid outputs, or to implement mechanisms that repeat certain parts of the simulation (such as repeating the prompting procedure) until valid output is obtained. Further, you may specify at which proportion of failed repetitions a whole condition will be excluded from the analysis.

Example

LLM ratings that do not fall within the specified range of admissible rating values (integers from 1 to 9 or from 0 to 100 depending on the condition) will be treated as missing values. We will report the proportion of such cases and exclude conditions if the proportion of missing values exceeds 20% based on a rule of thumb.

Answer:

6.5 How do you plan on interpreting the performance measures? (optional)

Explanation: You can optionally specify thresholds for drawing conclusions to reduce post-hoc flexibility in interpreting the results (e.g., what are considered 'acceptable' and 'unacceptable' levels of performance and what are 'relevant differences' between LLMs in terms of their performance to fulfill a particular task). Such thresholds are conventions rather than objective quality criteria; their main purpose is to commit to an interpretation before seeing the results, and they should be justified with respect to the intended use of the ratings. If specific comparisons between LLM configurations are planned (e.g., whether a smaller local model agrees with human ratings as well as a commercial model), specify them in advance together with the statistical test that will be used. Because both correlations share the same benchmark ratings, they are dependent and should be compared with tests for dependent correlations, for instance, as implemented in the R package `cocor` (Diedenhofen & Musch, 2015). In exploratory multiverse designs, the number of possible pairwise comparisons may be large, so that summarizing the overall pattern (e.g., by variance components or plots of all conditions) is often more useful than testing every pair.

Example

Regarding our primary performance measure, the correlation between human and LLM ratings, we define the following thresholds: an inter-rater reliability of $\widehat{IRR} \geq .70$ will be labeled acceptable, $\widehat{IRR} \geq .80$ adequate, and $\widehat{IRR} \geq .90$ (as obtained with GPT-4o by Martínez et al., 2024) excellent. These cutoffs follow common conventions for reliability coefficients, which were proposed mainly in the context of test–retest reliability (Groh, 2025; Matheson, 2019). We consider a certain method as performing better (or worse) than a threshold on the employed corpus of text if the MCSE 95% CI is above (below) this threshold. We consider method X as performing better than method Y on the employed corpus if the lower bound for the estimated inter-rater reliability of method X ($\widehat{IRR} - 1.96 \cdot \text{MCSE}$) is greater than the upper bound for method Y ($\widehat{IRR} + 1.96 \cdot \text{MCSE}$). Because the MCSE

only reflects the finite number of iterations, these conclusions do not generalize to new stimuli. To draw conclusions about comparable stimulus material, we use the 95% CI based on the averaged standard errors of the correlations between human-generated and LLM-generated ratings. If this interval is above (below) a threshold, we conclude that a method performs better (worse) than this threshold.

Answer:

7 Other

7.1 Which statistical software/packages do you plan to use?

Explanation: Likely, not all software used can be prespecified before conducting the LLM simulation. However, the main packages used for working with LLMs are usually known in advance and should be listed here, ideally with version numbers.

Example

We will use `R` version 4.6.0 (R Core Team, 2025) and the following R packages in their most recent versions: `ellmer` 0.4.0 (Wickham et al., 2025) for standardizing the API calls to the locally installed LLMs, which are run via Ollama 0.12.9 (https://ollama.com/), `tidyverse` 2.0.0 (Wickham et al., 2019) and `ggplot2` 4.0.3 (Wickham, 2016) packages for data analysis and visualization, and `gtheory` 0.1.2 (Moore, 2016) and lme4 2.0-1 (Bates et al., 2015) for conducting generalizability theory analyses.

Answer:

7.2 Which computational environment do you plan to use?

Explanation: Please specify the operating system and its version which you intend to use. If the study is performed on multiple machines or servers, provide information for each one of them, if possible.

Example

We will run the LLM simulation study on a Kubuntu 24.04 machine using an Nvidia GPU (RTX 5090) with the Cuda 13.0 environment. The complete output of `sessionInfo()` will be saved and reported in the supplementary materials.

Answer:

7.3 Which other steps will you undertake to make LLM simulation results reproducible? (optional)

Explanation: This can include sharing the code and full or intermediate results of the simulation in an open online repository. Additionally, this may include supplemental materials or interactive data visualizations, such as a shiny application.

Example

We will upload the fully reproducible simulation script and a data set containing all LLM ratings of valence and arousal as well as the correlation between LLM ratings and human ratings including confidence intervals to OSF.

Answer:

7.4 Is there anything else you want to preregister? (optional)

Explanation: For example, the answer could include the most likely obstacles in the repeated call of LLMs (API or local) or the simulation design, and plans to overcome them, or measures that increase the trust in the preregistration date (e.g., setting the seed based on a future event).

Example

No.

Answer:

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A Prompt Examples

A.1 Prompt Construction

Explanation: The appendix should provide examples for the prompt instructions and verbal materials for conducting the LLM simulation study. Extensive materials such as the complete corpus of text materials or all prompt versions may be stored and made available in separate files (e.g., CSV or text files).

Example

Each LLM prompt consists of five components that are concatenated in each iteration of the simulation:

1. An explanation of the task,
2. the range of responses allowed,
3. examples for multi-word expressions,
4. the target multi-word expression to be rated, and
5. a reminder for the range of responses allowed.

The examples below are based on Prompt 2 provided by Martínez et al. (2024) for ratings of valence. Ratings of arousal were analogous modifications of Prompt 3 (not provided here).

Example prompt with examples of low and high valence (original version of Prompt 2 in Martínez et al.):
Could you please rate how reading the following multi-word expression makes a person feel. Use a scale from 1 to 9, where 1 means very negative, bad and 9 means very positive, good. Examples of words that would get a rating of 1 are pedophile, AIDS, and wreck. Examples of words that would get a rating of 9 are vacation, fantastic, and laugh. The expression is: [insert expression here]. Only answer a number from 1 to 9. Please limit your answer to numbers.

Example prompt with only examples of low valence:
Could you please rate how reading the following multi-word expression makes a person feel. Use a scale from 1 to 9, where 1 means very negative, bad and 9 means very positive, good. Examples of words that would get a rating of 1 are pedophile, AIDS, and wreck. The expression is: [insert expression here]. Only answer a number from 1 to 9. Please limit your answer to numbers.

Examples for multi-word expressions are (taken from Martínez et al., 2024):

- hit rock bottom
- fall in love

- ice cream cone
- drug addict
- starry sky